

Tan Kee Xiang

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EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE (NUS)

Aug 2023 - Present

Bachelor of Science in Business Analytics

- GPA: 4.38, Second Class (Upper) Honours/ Honours (Distinction)
- Prospective Specialisation in Machine Learning based Analytics

NGEE ANN POLYTECHNIC

Apr 2018 - Apr 2021

Diploma with merit in Financial Informatics

- GPA: 3.86
- Directors List (2018-2020)
- Specialisation in Financial Analytics

EXPERIENCE

MINISTRY OF HOME AFFAIRS *Data Scientist Intern*

May 2025 - Dec 2025

- Built an LLM-driven NER and relationship extraction pipeline using GPT-4o, LangChain and Azure Document Intelligence to extract Cyber Threat Intelligence (CTI) entities and map into STIX-compliant structures.
- Designed a pre-processing workflow (OCR, PDF parsing, chunking and regex cleaning) and integrated extracted data into Neo4j for graph-based visualisation of threat actors, IOCs and TTP relationships, improving intelligence analysis.
- Developed big-data geospatial analytics workflows on Databricks using PySpark, Apache Sedona and Uber H3 to analyse vessel movements and maritime patterns.
- Applied H3 hierarchical indexing to improve spatial aggregation, ship trajectory modelling and maritime anomaly detection for operational investigations.

TOYOTA TSUSHO ASIA PACIFIC *Machine Learning Researcher*

Sep 2020 - Feb 2021

- Developed and evaluated ML models to predict supplier delivery delays using real-world supply chain data
- Executed large-scale data cleaning, feature engineering, and model evaluation across multiple algorithms
- Created Tableau dashboards to communicate insights on supplier capacity, order volatility and delivery performance

PROJECTS

National Cancer Centre Singapore, AI Engineer

Jan 2026 - Apr 2026

Agentic Collaborative Reasoning for cancer research

- Built a LangGraph multi-agent system enabling clinicians to query OMOP CDM healthcare data using natural language while running locally with Ollama to preserve patient data privacy
- Constructed a Human-in-the-Loop (HITL) workflow with clinician review checkpoints leveraging LangGraph interrupts, ensuring auditability before results were returned
- Enhanced accuracy of clinical concept mapping by replacing traditional lexical search with a vector-based similarity search engine using ClinicalBERT and FAISS within the OMOP domain

Judicial Analytics, Data Scientist

Jan 2026 - Apr 2026

Predicting defendant liability in Singapore corporate litigation using deep learning

- Developed a Hierarchical Attention Network (HAN) with SwiGLU to capture multi-level legal nuances
- Engineered a 3-stage GPT pipeline to transform PDFs into ML-ready JSON via LLM-as-a-Judge

P2P Loan Default Prediction System, Data Scientist

Jan 2025 - Apr 2025

Built different models prioritising a high recall rate to detect potential defaulters

- Processed 2.9 mil loan records using PySpark and leveraged PCA to reduce features (83 to 62)
- Addressed class imbalances using SMOTE and ENN; trained LR, RF, SVM, and GBT classifiers

TECHNICAL SKILLS

- Programming Languages: Python, R, Java, C#, Javascript, HTML, CSS
- AI & Machine Learning: Predictive Modelling, Supervised and Unsupervised Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering, Feature Engineering, Model Evaluation
- Libraries & Frameworks: LangChain, LangGraph, Scikit-learn, Pandas, NumPy, Matplotlib, MLflow
- Data Engineering: ETL Pipelines, Apache Airflow
- Big Data & Cloud: Databricks, Apache Spark (PySpark), Apache Sedona, AWS
- Databases: SQL Database, Firebase, FAISS, Neo4j
- Data Visualisation & BI: Tableau, Excel
- Development Tools: Docker, Git, GitHub, Jupyter Notebook, VS Code